

Selecting Elements in JavaScript

When working with the DOM (Document Object Model), selecting elements is often the first step to manipulate them. JavaScript provides multiple methods to select HTML elements based on their ID, class, tag name, or CSS selector.

1. `getElementById`

Selects a single element by its unique `id`.

2. `getElementsByClassName`

Returns a live `HTMLCollection` of all elements with the specified class name.

3. `getElementsByTagName`

Returns a live `HTMLCollection` of all elements with the specified tag name (e.g., `div`, `p`, `h1`, etc.).

4. `querySelector`

Returns the first element that matches a specified CSS selector.

5. `querySelectorAll`

Returns a static `NodeList` of all elements that match a specified CSS selector.

These methods allow you to access and manipulate elements dynamically. We'll explore each of them in detail with examples next.